CSE 717 — Information Security

Topic 3: Symmetric vs Asymmetric & Block vs Stream Ciphers

Complete Past Paper Solutions: 2020–2024

Source: general security knowledge (no-source cheat sheet — Lc#1 pp.25–28 and Lc#7 p.12 are title-only slides with no body text)

Every Topic-3 question 2020–2024 included. Zero omissions.

Contents

Quick Reference — Cheat Sheet

No-source cheat sheet — like Topics #13 / #14

None of Lc#1–11 contain body text on: essential ingredients of a symmetric cipher, block vs stream cipher, classical Feistel structure, or public-key cryptosystem elements/roles. The slides only show title pages ("How Cryptography Works with Public and Private Key?", "Why Symmetric Algorithms Are Less Vulnerable"). Everything below is standard cryptography knowledge (Stallings) memorized as a cheat sheet.

1. Essential Ingredients of a Symmetric Cipher (5 components)

1. **Plaintext:** the original, readable message/data fed into the algorithm.
2. **Encryption algorithm:** performs substitutions/transformations on the plaintext.
3. **Secret key:** an independent value that controls the exact substitutions/transformations — both sender and receiver must share it via a secure channel.
4. **Ciphertext:** the scrambled output, depends on the plaintext *and* the key.
5. **Decryption algorithm:** the encryption algorithm run in reverse, using the same secret key, to recover the plaintext.

2. Block Cipher vs Stream Cipher (one-line difference)

- **Block cipher:** encrypts a fixed-size **block** of plaintext (e.g. 64/128 bits) at a time, using the same key for each block (e.g. DES, AES).
- **Stream cipher:** encrypts plaintext **one bit/byte at a time**, combining it with a pseudorandom keystream (typically via XOR) as the data streams in (e.g. RC4).
- **Memory aid:** block = "batch processing", stream = "one unit at a time, on the fly".

3. Classical Feistel Cipher Structure

A Feistel network splits each plaintext block into two halves L_i, R_i and applies n rounds. In each round i:

$$L_i = R_{i-1}, \qquad R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

where F is a round function and K_i is the round subkey (derived from the main key via a key schedule).

- Decryption uses the *same* structure with subkeys applied in **reverse order** — the same hardware/software can do both encryption and decryption.
- This is the structure underlying **DES** (see Topic #13 cheat sheet for the full diagram with f-function, XOR, and swap boxes per round).

4. Public-Key Cryptosystem: Elements & Key Roles

Six elements of a public-key cryptosystem:

1. **Plaintext** — the readable input message.
2. **Encryption algorithm** — performs transformations on the plaintext.
3. **Public key & private key** — a mathematically linked pair; one for encryption/verification, the other for decryption/signing.
4. **Ciphertext** — the scrambled message produced by the algorithm + key.
5. **Decryption algorithm** — accepts the ciphertext and the matching key to recover the plaintext.

Roles of the keys:

- **Public key:** freely distributed; used by anyone to *encrypt* a message for the owner, or to *verify* the owner's digital signature.
- **Private key:** kept secret by the owner; used to *decrypt* messages encrypted with the matching public key, or to *create/sign* a digital signature.

2024 — Q1(a): Symmetric vs Asymmetric Crypto (3 marks) — cross-reference

Question — 2024 Q1(a)

Briefly describe symmetric and asymmetric cryptography with an example. Point out the reason why asymmetric cryptography is useful for e-commerce. [3]

Answer — already fully solved under Topic 2

This question is bundled with Topic 2 (Security Fundamentals) in 2024 and is fully answered in `SecurityFundamentals_Solutions.pdf`, Section "2024 — Q1(a,b)". In short: **symmetric** crypto uses one shared secret key for encryption *and* decryption (fast, e.g. AES); **asymmetric** crypto uses a public/private key pair (e.g. RSA) — useful for e-commerce because it removes the need for a pre-shared secret and enables secure key exchange + digital signatures over an insecure channel.

2022 — Q2(a): Essential Ingredients of a Symmetric Cipher (3 marks)

Question — 2022 Q2(a)

What are the essential ingredients of a symmetric cipher? [3]

Answer — 2022 Q2(a)

A symmetric cipher has **five** essential ingredients:

1. **Plaintext** — the original message.
2. **Encryption algorithm** — performs substitutions/transformations on the plaintext, guided by the key.
3. **Secret key** — shared in advance between sender and receiver via a secure channel; the exact transformations depend on this key.
4. **Ciphertext** — the scrambled output, a function of both the plaintext and the key.
5. **Decryption algorithm** — the encryption algorithm in reverse, using the same secret key, to recover the original plaintext.

Exam tip: draw the standard block diagram — Plaintext → [Encryption Algorithm ← Secret Key] → Ciphertext → [Decryption Algorithm ← Secret Key] → Plaintext — if a diagram is requested.

2022 — Q2(c): Public-Key Cryptosystem — Elements & Key Roles (2+2 marks)

Question — 2022 Q2(c)

What are the principal elements of a public-key cryptosystem, and what are the roles of the public and the private key? [2+2]

Answer — Elements (2 marks)

A public-key cryptosystem has six elements: **plaintext**, **encryption algorithm**, **public key**, **private key**, **ciphertext**, and **decryption algorithm**. The key pair is mathematically related such that what one key encrypts, only the other can decrypt.

Answer — Roles of the public and private key (2 marks)

- **Public key:** published openly. Anyone can use it to **encrypt** a message intended for the key-pair owner, or to **verify** a digital signature claimed to be from the owner.
- **Private key:** kept secret by the owner. Used to **decrypt** messages that were encrypted with the matching public key, or to **generate/sign** a digital signature.

Memory aid: "Public locks (encrypts/verifies), Private unlocks (decrypts/signs)."

2020 — Group-B Q5(b): Classical Feistel Cipher Structure (3 marks)

Question — 2020 Q5(b)

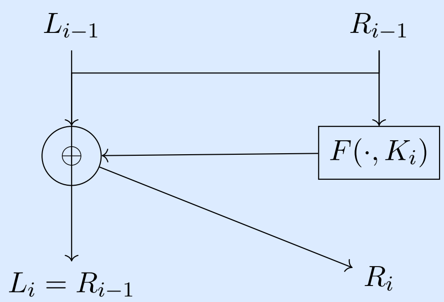
Draw the classical Feistel cipher structure for the symmetric block encryption algorithm. [3]

Answer — 2020 Q5(b)

A single Feistel round on a $2w$-bit block split into halves L_{i-1} (left) and R_{i-1} (right):

$$L_i = R_{i-1}, \qquad R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$$

Diagram (one round):



Key points to write:

- Each round: swap the halves, and XOR the right half (after passing through round function F with subkey K_i) into the left half.
- Repeated for n rounds (e.g. DES uses 16 rounds).
- **Decryption** reuses the exact same structure, just with the subkeys $K_n, K_{n-1}, \ldots, K_1$ applied in **reverse order** — a major practical advantage (same circuit/code for both directions).

2020 — Group-B Q5(c): Block Cipher vs Stream Cipher (1 mark)

Question — 2020 Q5(c)

What are the differences between a block cipher and a stream cipher? [1]

Answer — 2020 Q5(c)

Block Cipher	Stream Cipher
Encrypts a fixed-size block of bits (e.g. 64/128 bits) at a time	Encrypts one bit/byte at a time as data streams in
Same key reused per block (with a mode of operation)	Combines plaintext with a pseudorandom keystream, typically via XOR
Examples: DES, AES	Example: RC4

One sentence is enough for 1 mark: "A block cipher encrypts fixed-size blocks of data at once, while a stream cipher encrypts data bit-by-bit (or byte-by-byte) continuously."

Exam Strategy Summary

High-Yield Checklist for Symmetric/Asymmetric & Block/Stream Ciphers

1. **5 ingredients of a symmetric cipher** — plaintext, algorithm, secret key, ciphertext, decryption algorithm (1 mark each if listed, full marks if briefly explained).
2. **Block vs stream cipher** — one-line: "fixed block at once" vs "bit/byte-by-byte continuously". Quick 1-mark win, asked in 2020.
3. **Classical Feistel structure** — $L_i = R_{i-1}$, $R_i = L_{i-1} \oplus F(R_{i-1}, K_i)$, plus "same structure decrypts with reversed subkeys". Shared with DES cheat sheet (#13) — learn once, use twice.
4. **Public-key cryptosystem** — 6 elements (plaintext, algorithm, public key, private key, ciphertext, decryption algorithm) + key roles ("public locks, private unlocks").
5. **Symmetric vs asymmetric (overlap with Topic 2)** — one shared key vs key pair; asymmetric solves key-distribution + enables signatures.
6. This whole topic is **definition/diagram recall**, no numericals — a 15-20 minute cheat-sheet pass should cover it cold.